

NORTHWIND SALES PERFORMANCE DASHBOARD

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Course: BBA – Digital Marketing & Business Analytics

Tool Used: Microsoft Power BI Desktop

1.Introduction

For this project, I created an **interactive Northwind Sales Dashboard using Microsoft Power BI**. The main purpose of the project was to analyse sales data and present it through simple charts, KPI cards and filters.

The dashboard helps to understand important areas such as **sales, orders, customers, quantity sold, freight, product categories and employee-wise orders**. It also allows users to interact with the data using filters.

2.Objective:

The main objectives of this project were:

- To analyse the Northwind sales data.
- To create important sales KPIs using DAX.
- To understand monthly order trends.
- To compare sales between different categories.
- To analyse employee-wise orders.
- To create an interactive Power BI dashboard.
- To use filters, drill-through and tooltips for better analysis.

3. Dataset Description

The Northwind dataset contains information related to the company's sales activities.

The main tables used in this project were:

- **Categories** – Contains product category information.
- **Customers** – Contains customer details.

- **Employees** – Contains employee information.
- **Orders** – Contains order information.
- **Order Details** – Contains product quantity and sales details.

These tables were used together to create the required calculations and visualizations.

4. Data Preparation

I imported the Northwind dataset into **Power BI Desktop** and checked the available tables and fields.

The main preparation steps were:

- Checking the data types.
- Checking date and numerical fields.
- Connecting the related tables.
- Creating the required DAX measures.
- Checking the calculations before creating the visuals.

This preparation helped ensure that the data was ready for analysis.

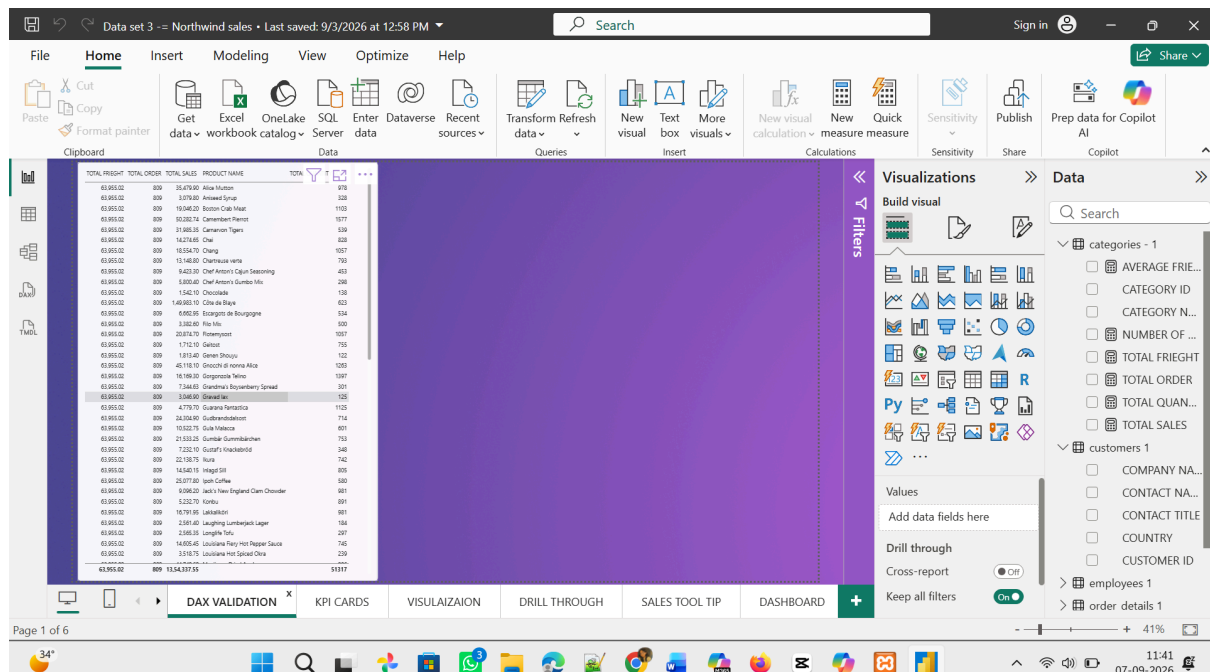
5. DAX Validation

I created a separate **DAX Validation** page to check whether the calculations were working correctly.

The validation included measures such as:

- Total Orders
- Total Sales
- Total Quantity
- Total Freight

The calculated values were then used in the KPI cards and other visuals in the report.

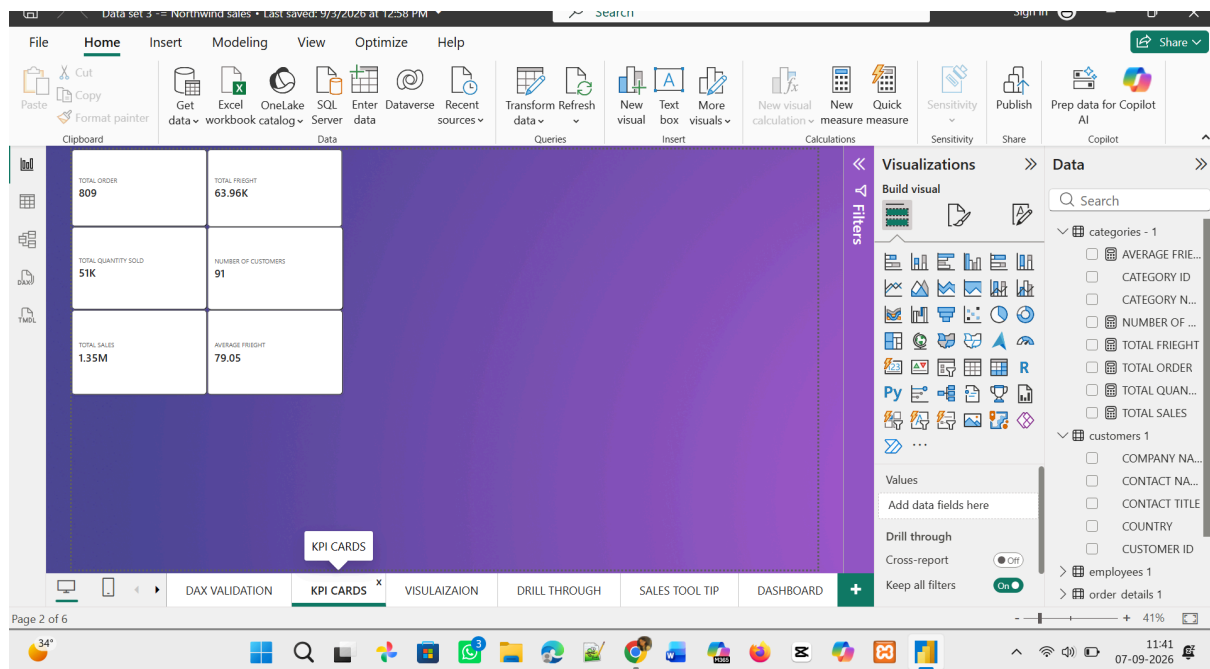


6. KPI Cards

I created **six KPI cards** to show the main sales information in a simple format.

KPI	Value
Total Orders	809
Number of Customers	91
Total Freight	63.96K
Total Sales	1.35M
Total Quantity Sold	51K
Average Freight	79.05

These KPI cards provide a quick summary of the overall sales performance.



7. Visualizations

I created different visualizations to understand the Northwind sales data.

Total Orders by Month

A **line chart** was used to show the monthly order trend. It helps identify changes in the number of orders throughout the year.

Total Orders by Employee

A **bar chart** was used to compare the number of orders associated with different employees. It helps understand the difference in order volume between employees.

Total Sales by Category

A **column chart** was used to compare sales across different product categories. From the visual, **Beverages** show the highest sales among the categories displayed.

These visualizations make the data easier to compare and understand.

8. KPI Cards – Explanation

The KPI cards were created to present the most important business figures at a glance.

Total Orders – 809:

Shows the total number of orders in the dataset.

Number of Customers – 91:

Shows the number of customers included in the analysis.

Total Freight – 63.96K:

Shows the total freight amount associated with the orders.

Total Sales – 1.35M:

Shows the total sales generated from the Northwind data.

Total Quantity Sold – 51K:

Shows the total quantity of products sold.

Average Freight – 79.05:

Shows the average freight value of the orders.

Purpose of KPI Cards

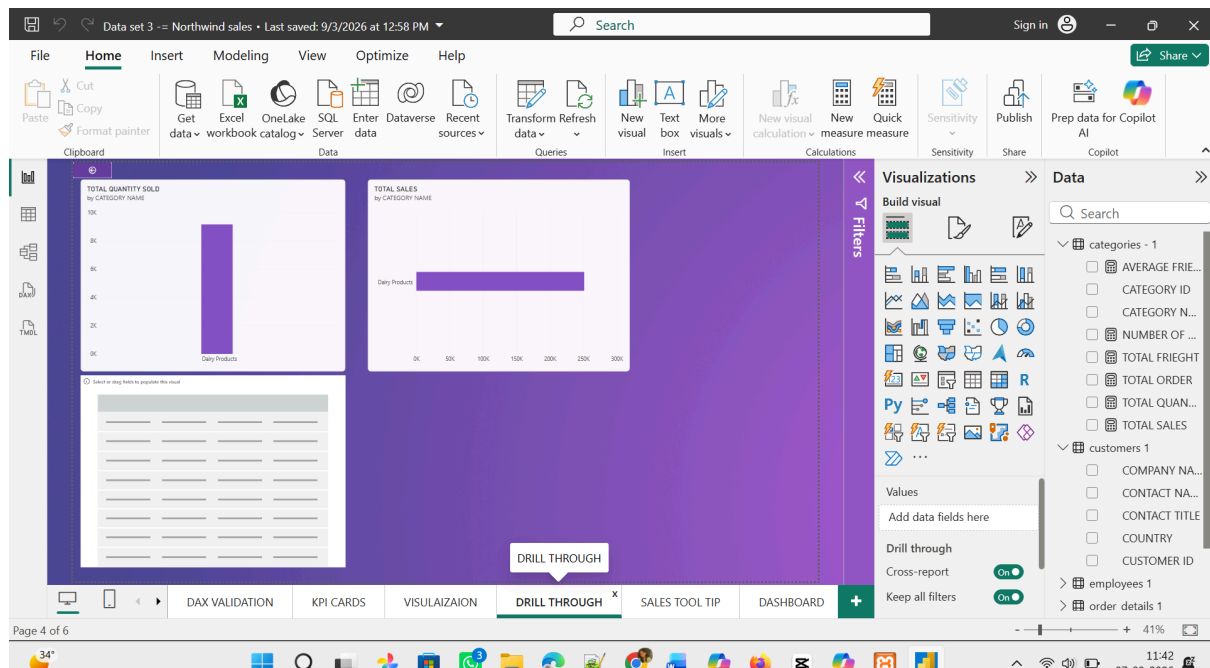
I used KPI cards because they provide a **quick overview of the business performance**. The user can understand the main figures without having to analyse each individual chart.

9. Drill Through

I created a separate **Drill Through** page for detailed analysis.

The drill-through feature allows the user to move from a summary-level visual to more detailed information based on the selected data.

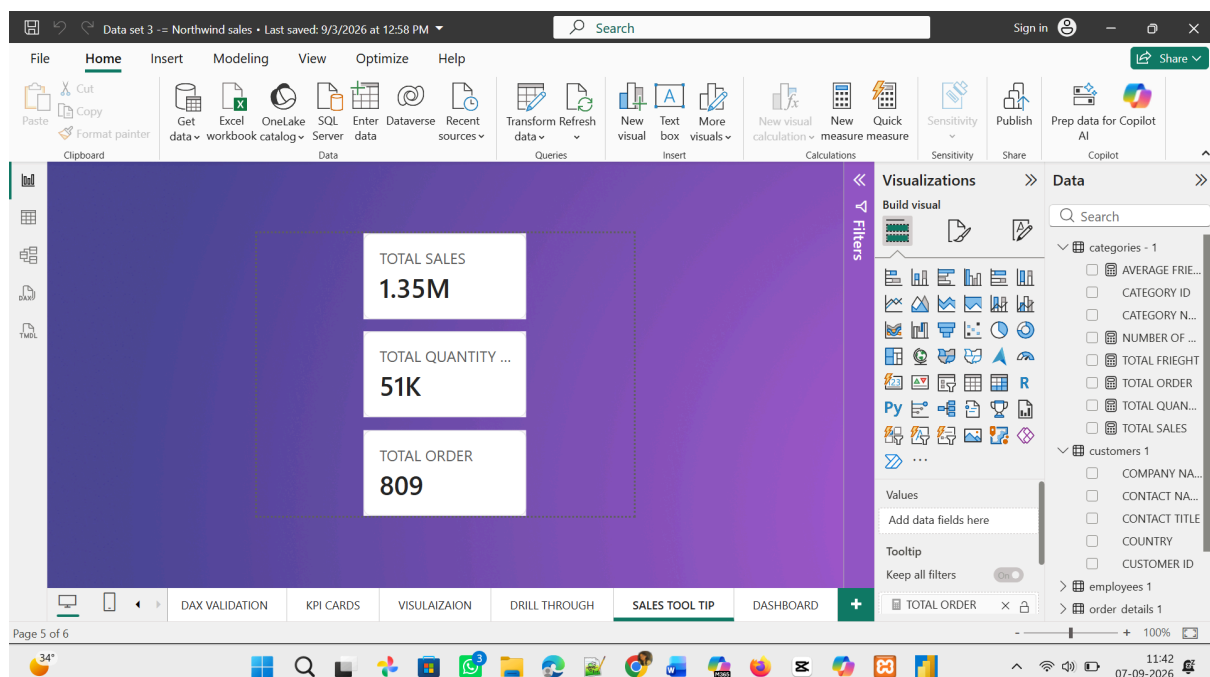
This makes the report more interactive and helps in analysing specific categories or records in greater detail.



10. Sales ToolTip

I created a **Sales ToolTip** page to provide additional information when the user interacts with a visual.

The tooltip helps display extra information without adding too many details to the main dashboard. This keeps the dashboard simple while still providing additional information when required.



11. Dashboard & Filters

The final dashboard combines the important **KPI cards and visualizations** in one page.

The dashboard includes:

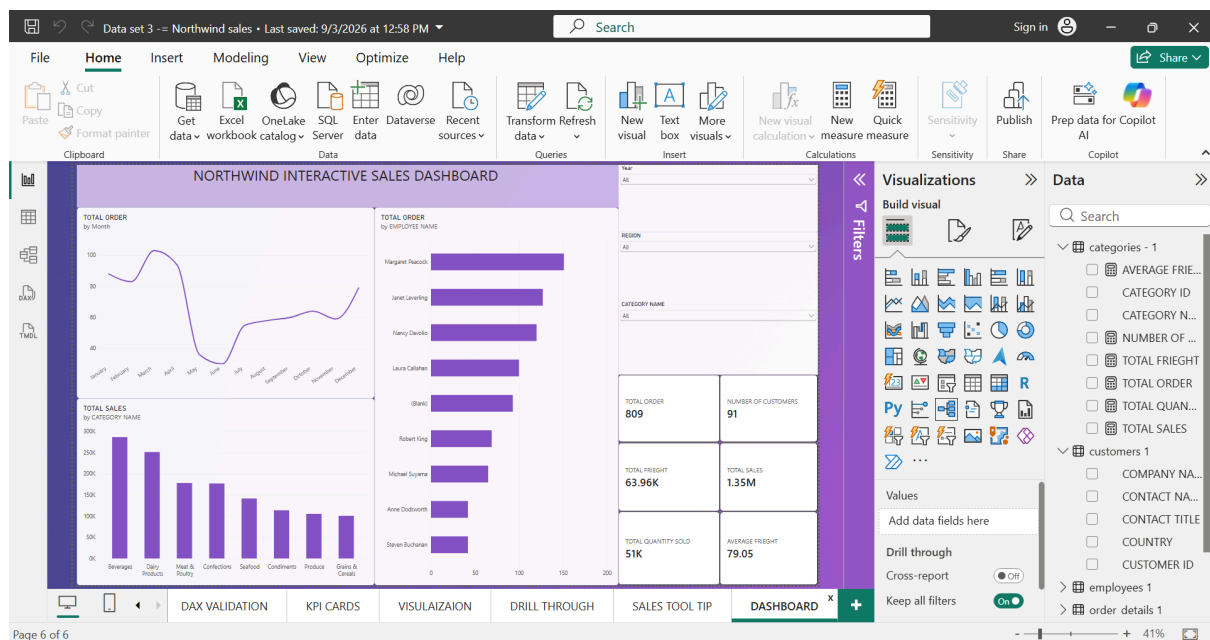
- Total Orders by Month
- Total Orders by Employee
- Total Sales by Category
- KPI Cards
- Interactive Filters

I added three main filters:

- **Year**
- **Region**
- **Category Name**

These filters allow the user to select specific data and analyse the dashboard based on their requirement.

For example, selecting a particular category can help analyse the sales performance of that product category. Similarly, the Year and Region filters help narrow down the analysis.



12. Key Findings

From the completed dashboard, I observed that:

- Total sales are **1.35M**.
- There are **809 total orders**.
- The dataset contains **91 customers**.
- Around **51K quantity** was sold.
- Total freight is **63.96K**.
- Average freight is **79.05**.
- **Beverages** have the highest sales among the categories shown.
- Order volume changes across different months.
- Employee-wise order volumes are different.
- The filters make it easier to analyse the data based on Year, Region and Category.

13. Conclusion

This project helped me understand how **Power BI can be used to convert sales data into meaningful business information**.

Through this activity, I learned how to prepare data, create DAX measures, validate calculations, create KPI cards and design different visualizations.

The final Northwind dashboard provides a simple way to understand **sales, orders, customers, product categories and employee performance**. The use of filters, drill-through and tooltips also made the report more interactive.

Overall, this project improved my practical knowledge of **Power BI, DAX, data visualization and business analysis**.

14. Reference

- Microsoft Power BI Desktop
- Northwind Sales Dataset
- Microsoft Power BI Documentation